

# THE ZERO PARADOX

## ZP-I: Inside Zero

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This document establishes Theorem T-IZ (Inside Zero): every maximal ascending chain in the Zero Paradox framework that strictly ascends at every step — so that its 2-adic valuation grows without bound — is a Cauchy sequence that converges, in the 2-adic metric, to 0. No new axioms are required for what it PROVES, which is that convergence, together with the separate fact that anything filling the bottom role IS the bottom of its own semilattice. Reading the 2-adic limit as the thing filling that role is a modelling commitment, not a theorem: the two live in different types, and  $Q_2$  carries no join, so the role condition is not statable of the limit at all — they are distinct MEMBERS of the bottom family, not one object, and that identity is retired as ill-typed (MC-1). Reading the bottom so reached as a NEW one rather than the one already there is a further commitment (C-DA2 in ZP-E), and in the 2-adic chart the arc returns to the SAME 0 (snap\_arc\_z2\_loop). T-IZ extends T-SNAP: where T-SNAP fixes the shape of the first transition  $\perp \rightarrow \epsilon_0$ , and not that it is taken, T-IZ establishes, for a chain that does take every step, the full trajectory — ascent through  $\omega$  state changes and convergence back to 0 — as a single Cauchy sequence result.

The key insight: the no-top property is not an obstacle to T-IZ. It is what makes the ascent AVAILABLE — there is always a strictly greater element, so no chain halts for want of anywhere to go. (It does NOT follow that the limit lies outside L; whether a limit escapes its carrier is a property of the particular lattice — the ordinals under max have no top, and the chain  $n \mapsto n$  has least upper bound  $\omega$  inside them.) What drives the 2-adic valuation  $v_2(S_n) \rightarrow \infty$  is that the chain actually takes those steps (IsStrictStateSequence, the hypothesis  $h\_strict\_from\_r1\_t3$  binds); no-top alone does not, and the NO-GO gauge in §Ib exhibits a no-top lattice holding a chain that never moves. Unbounded ascent is exactly the Cauchy convergence condition  $\|S_n\|_2 \rightarrow 0$ . The chain approaches the 2-adic depth of zero by going deeper into the p-adic structure — not by reversing direction. Read as reaching maximum complexity, DA-1 and T-SNAP apply and the framework calls the result a successor null  $\perp'$  — both of those steps are readings, not consequences of the convergence.

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## Section I: Room to Ascend — No Top Element and Ordinal Unboundedness

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### I. The No-Top Property: What It Supplies

The no-top property (Lean: HasNoTop, ZeroParadox/Order/Lattice.lean) says the join-semilattice  $(L, \vee, \perp)$  has no top element: there is no  $T \in L$  with  $x \leq T$  for all  $x$ .  $\triangleup$  This is NOT ZP-A's R1, which is the NO-SUBTRACTION restriction (§I of the ZP-B section below); one label over two propositions is a citation nothing can check. What no-top supplies is ROOM: a chain never halts for want of a strictly greater element. The ascent itself is IsStrictStateSequence — see the NO-GO gauge in §Ib. Reading the limit as occupying the bottom role is a commitment, and calling the occupant a successor null is C-DA2, a further one.

The reasoning is direct. An ascending chain  $(S_n)_{n < \omega}$  in  $L$  is a sequence satisfying  $S_n \leq S_{n+1}$  for all  $n$  (ZP-A T3 — monotonicity). Because  $L$  has no top element, for every  $S_N$  there EXISTS an  $S$  with  $S_N < S$  — so no chain is ever forced to stabilise for want of somewhere to go.  $\triangle!$  That is availability, not occurrence: a CONSTANT chain is still a legitimate state sequence in a no-top lattice (NO-GO gauge, §Ib). That this chain is strictly ascending and unbounded is the separate hypothesis `IsStrictStateSequence`, and that is the one T-IZ consumes.

In the 2-adic model (ZP-B), each element  $S_n$  corresponds to an element of  $Q_2$  with 2-adic valuation  $v_2(S_n)$ . Strict ascent in  $L$  corresponds to increasing 2-adic valuation depth. Because the chain is unbounded,  $v_2(S_n) \rightarrow \infty$ . This follows from the strict-ascent hypothesis, not from the absence of a top element: no-top leaves room for the ascent, and `IsStrictStateSequence` is that the chain takes it.

## II. Ordinal Index Replaces Clock Time

The state sequence is indexed by ordinals:  $(S_\alpha)_{\alpha < \omega}$ . The parameter  $\omega$  is not a clock time and not a top bound. It is the ordinal index of the transition — the label for when the chain has completed  $\omega$  state changes. The chain in  $L$  is genuinely unbounded; the ordinal  $\omega$  is not a member of the sequence and not a ceiling on it.

This replaces the informal "time" language that sometimes accompanies descriptions of the Binary Snap. In the Zero Paradox, "time" is the index of state changes. What fixes the direction of that index is monotonicity (ZP-A T3) and irreversibility (ZP-B C3). Neither is a clock. The successor null does not appear at a future clock time — it is reached at the limit ordinal  $\omega$ , after  $\omega$  state changes have occurred. (Reached, and identified by its role. Whether it is a NEW bottom or the same one is the commitment, not the timing.)

Remark R-I.1:  $\omega$  is the first infinite ordinal — the smallest ordinal greater than every natural number. An ascending chain indexed by  $\omega$  is a countable sequence with no last element in  $L$ . The no-top property guarantees only that no finite stage is a CEILING; that a given chain reaches through all of them is the strict-ascent hypothesis. The transition at  $\omega$  is not a step within  $L$ .  $\triangle!$  Whether its limit lies OUTSIDE  $L$  is a property of the particular lattice and not a consequence of no-top.

### Key Result

Strict ascent (`IsStrictStateSequence`) drives  $v_2(S_n) \rightarrow \infty$ ; the no-top property supplies the room for it, and is not a hypothesis OF THAT THEOREM (it is a per-carrier assumption in its own right — `nat_has_no_top` is a theorem, and `zpa_bot_not_greatest` binds it explicitly). Strict valuation growth ( $v_2(S(n)) \geq n$ ) is Lean-derived from `IsDepthChain` and `IsStrictStateSequence` — Lean: `h_strict_from_r1_t3` (§Ib). R-IZ-A closed.

$$\|S_n\|_2 = 2^{-v_2(S_n)} \rightarrow 0 \text{ (Cauchy condition).}$$

The no-top property removes the ceiling; `IsStrictStateSequence` is that the chain climbs. Of those two only `IsStrictStateSequence` appears in `h_strict_from_r1_t3`'s binders — alongside `IsDepthChain`, which is not a lattice condition but the bridge to the 2-adic valuation.

## Section II: The Two Paths to $P_0$

The approach from inside can be traced along two parallel paths: one topological (through  $Q_2$  and the 2-adic norm), one informational (through ZP-C L-INF and the Kolmogorov complexity threshold  $P_0$ ). Both paths converge on the same condition ( $P_0$  satisfied at  $\omega$ ). They are not alternatives — they are two descriptions of the same structure.

## A. Topological Path — Cauchy Convergence in $Q_2$

The 2-adic norm on  $Q_2$  is defined by:  $\|x\|_2 = 2^{-v_2(x)}$ , where  $v_2(x)$  is the 2-adic valuation of  $x$ . In particular,  $v_2(0) = \infty$ , so  $\|0\|_2 = 0$  — the null element is the element of infinite 2-adic depth. As the ascending chain has  $v_2(S_n) \rightarrow \infty$  (forced by strict ascent, with `IsDepthChain` tying the valuation to the depth index), we have  $\|S_n\|_2 \rightarrow 0$  by the norm-valuation identity above.

In a normed group,  $\|S_n\|_2 \rightarrow 0$  already gives  $S_n \rightarrow 0$ : the limit is EXHIBITED as 0, not obtained from a Cauchy criterion, so no completeness is consumed at this step. Completeness of  $Q_2$  is what the CONVERSE direction needs — Cauchy implies some limit exists — and T-IZ never takes it. A convergent sequence is automatically Cauchy. The ascending chain is therefore a Cauchy sequence converging to 0 — the 2-adic limit of the chain is the null element.

### Lemma T-IZ-A — Cauchy Convergence (Proved in Lean)

Let  $S : \mathbb{N} \rightarrow Q_2$  be a sequence satisfying  $\|S(n)\|_2 \leq 2^{-n}$  for all  $n \in \mathbb{N}$ . Then:

(1) The norms  $\|S(n)\|_2 \rightarrow 0$  (squeeze between 0 and the geometric sequence  $2^{-n}$ , both tending to 0).

(2)  $S(n) \rightarrow 0$  in  $Q_2$  (norm  $\rightarrow 0$  iff sequence  $\rightarrow 0$  in a normed group).

Lean: `t_iz_cauchy` — proved (sorry-free) in `SemilatticeInstance.lean`; footprint carries `Classical.choice` inherited from Mathlib's p-adic analysis (see § III). This is the topological core of T-IZ.

The geometry of the inside approach is the following: elements of  $Q_2$  are arranged by their 2-adic valuation depth. Zero is the element of infinite depth — the deepest point. The ascending chain moves into greater and greater depth as  $n \rightarrow \infty$ , approaching the depth of zero without ever reversing. The chain does not turn around and head back to 0. It descends into 0 by going deeper.

Remark R-II.1: The condition  $\|S(n)\|_2 \leq 2^{-n}$  is equivalent to  $v_2(S(n)) \geq n$ . It asserts that the 2-adic valuation of  $S(n)$  is at least  $n$  — meaning  $S(n)$  is divisible by  $2^n$  in the 2-adic sense. As  $n \rightarrow \infty$ , divisibility by arbitrarily large powers of 2 forces  $\|S(n)\|_2 \rightarrow 0$  and therefore  $S(n) \rightarrow 0$ . This is the formal content of the "chain approaching the 2-adic depth of zero by forward motion."

Remark R-IZ-A — Closure of the valuation growth hypothesis: The strict growth condition  $v_2(S(n)) \geq n$  was previously treated as a construction-level assumption stronger than the proved result `t_iz_valuation_unbounded` ( $\sup v_2(S(n)) = \infty$ ). It is now Lean-derived. Theorem `h_strict_from_r1_t3` (`SemilatticeInstance.lean` §Ib) proves that any  $Q_2$  chain satisfying the `IsDepthChain` modeling commitment — meaning its 2-adic valuations track a strict  $\mathbb{N}$ -depth-index sequence — inherits strict valuation growth from `IsDepthChain` and `IsStrictStateSequence`. `IsDepthChain` ( $\forall n, v_2(S(n)) = \text{depths}(n)$ ) is the remaining modeling commitment: it asserts that 2-adic depth tracks the lattice depth index. This is a structural feature of the embedding, not a consequence of the abstract axioms. With `IsDepthChain` in place, R-IZ-A is formally closed.

## B. Informational Path — The Valuation-Complexity Bridge

ZP-C L-INF establishes that the surprisal  $I(n) = n$  at ball-hierarchy depth  $n$  is unbounded.  $\perp$  corresponds to the limit point  $0 \in Q_2$  — the limit of the binary ball hierarchy at infinite depth. The depth-surprisal correspondence (ZP-C D4) gives the informational content of the ascending chain: as  $v_2(S_n) \rightarrow \infty$ , the surprisal  $I(n) \rightarrow \infty$  without bound.

In the framework's binary construction — binary alphabet, ball-hierarchy depth equalling surprisal (ZP-C D4), and Kolmogorov complexity measuring descriptive incompressibility — 2-adic valuation depth and Kolmogorov complexity are measuring the same structure from two sides. The topological path traces depth-in- $Q_2$ ; the informational path traces descriptive incompressibility. As both grow without bound, they converge on the same condition: the incompressibility threshold  $P_0$  (ZP-C D1).

### Bridge Claim — Valuation-Complexity Bridge

Claim:  $v_2(S_n) \rightarrow \infty \Rightarrow K(S_n | n) / |S_n| \rightarrow 1$ .

In the binary framework's construction, the 2-adic valuation depth (topological) and Kolmogorov complexity (informational) are two descriptions of the same structure. At the Cauchy limit, both converge on  $P_0$ ; the incompressibility threshold  $K(c_1 | n) / |c_1| = 1$  (ZP-C D1).

Lean scope: Kolmogorov complexity  $K$  is uncomputable and absent from standard proof libraries. Bridge is Outside Lean Scope — same category as DA-1 Path 3 (ZP-C D1 + AIT) in ZP-E. The topological core (§ A above) is proved in Lean, carrying Classical.choice from Mathlib's p-adic analysis (§ III); the bridge follows the ZP-E informal argument. See ZP-E § IV for the full DA-1 Path 3 treatment that the bridge extends.

Remark R-II.2: The formal spine of T-IZ is Steps 1 and 6. Step 1 (Cauchy convergence to 0 in  $Q_2$ ,  $t\_iz\_cauchy$ ) is proved in Lean carrying Classical.choice from Mathlib's p-adic analysis. Step 6 (anything filling the  $\perp$  ROLE IS that lattice's  $\perp$ ) is proved axiom-free in Lean via  $t\_iz\_limit\_is\_new\_null$  — which establishes the role identification ONLY and must not be cited as a witness for the successor reading ( $t\_iz\_limit\_is\_new\_null$ , declared in ZeroParadox/Valuation/SemilatticeInstance.lean and indexed in ZeroParadox/Order/SnapCannotBe.lean). • The two steps are not joined. Step 1 is about  $0 \in Q_2$ ; Step 6 is about a terminal in a semilattice  $L'$ , and takes the role property as its hypothesis  $h\_role$ .  $t\_iz\_complete$  conjoins them without identifying the two objects, and the identification is not merely unproved —  $Q_2$  carries no join, so the role condition cannot be stated of the limit at all. That junction is not an open gap: the 0 of  $Q_2$  and the algebraic  $\perp$  are distinct MEMBERS of the bottom family, and the cross-category identity is retired as ill-typed rather than outstanding (MC-1, CLAIMS.md). Steps 2–5 describe the original ZP-E informational argument connecting 2-adic depth to Kolmogorov complexity and DA-1 Path 3. Since DA-1 is closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement — without Kolmogorov complexity — Steps 2–5 are informational context, not a proof dependency. The bridge is retained as historical motivation: it documents why the framework's informational and topological layers converge at  $P_0$ .

## Section III: Theorem T-IZ — Inside Zero

## Theorem T-IZ — Inside Zero

Statement: Every maximal ascending chain  $(S_n)_{n < \omega}$  in the Zero Paradox framework that strictly ascends at every step, so that its 2-adic valuation grows without bound, is a Cauchy sequence that converges, in the 2-adic metric, to 0.

Reading that limit as an OCCUPANT of the bottom role is a modelling commitment and not part of the statement —  $Q_2$  carries no join, so the role condition is not statable of it. Reading that occupant as a SUCCESSOR null is C-DA2, a further commitment.

Formal hypotheses:  $S : \mathbb{N} \rightarrow Q_2$ , with  $S(0) = \perp$  (CC-1),  $S(n) \leq S(n+1)$  (T3 monotonicity), and  $v_2(S(n)) \geq n$  for all  $n$  (derived from `IsDepthChain` + `IsStrictStateSequence` — see Remark R-IZ-A, §Ib).

Conclusion:  $S(n) \rightarrow 0$  in  $Q_2$ . That is the whole of what is proved here. The framework then READS the limit as satisfying the bottom role — a step stated in a different type and not carried by T-IZ — and DA-2 supplies the one-directional fact that anything satisfying that role IS the bottom already present. On that reading  $P_0$  is satisfied, DA-1 fires and T-SNAP fires. DA-2 licenses reading that occupant as  $\perp'$ , the successor null for the next instantiation — a modelling commitment, since nothing here produces a SECOND bottom.

## I. The Six-Step Proof

The proof of T-IZ follows six steps, corresponding to the proof obligation table:

- Step 1 — Cauchy convergence: The ascending chain has  $\|S(n)\|_2 \leq 2^{-n}$  (from  $v_2(S(n)) \geq n$  — Lean-derived via `h_strict_from_r1_t3` given `IsDepthChain`; R-IZ-A closed). By T-IZ-A (§ II.A),  $S(n) \rightarrow 0$  in  $Q_2$ . Proved in Lean: `t_iz_cauchy` (carries `Classical.choice` from Mathlib p-adic analysis, § III). ✓
- Step 2 — Valuation-complexity bridge (informational context): As  $v_2(S(n)) \rightarrow \infty$ ,  $K(S(n)|n)/|S(n)| \rightarrow 1$ . Original informational route to DA-1 Path 3. Not a proof dependency for T-IZ — DA-1 is closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement. Retained as motivational context connecting the topological and informational layers.
- Step 3 —  $P_0$  is satisfied at the limit: ZP-C D1 gives  $K(c_1|n)/|c_1| = 1$  at the limit. The configuration is algorithmically incompressible. ZP-C D1 applies.
- Occurrence fence. T-SNAP fixes the SHAPE of each step. It does not establish that any step is taken: `tsnap_holds_but_nothing_moves` exhibits a model in which T-SNAP holds and nothing moves. Throughout this document, "fires" narrates the commitment that instantiation occurs - before this note as well as after it - not a consequence of the theorem.
- Step 4 — DA-1 fires: A configuration at  $P_0$  is a live execution event — not a static description. DA-1 (ZP-E) applies, with the same three-path argument as in ZP-E § IV. The TrackedOutput formal core (DP-2, Snap.lean § VI) establishes the machine-state transition.
- Step 5 — T-SNAP fires: DA-1 establishes instantiation = execution. T-SNAP (ZP-E) gives  $\perp \vee \epsilon_0 = \epsilon_0$ . Lean: `t_snap_derived`, proved axiom-free in Snap.lean. ✓ — the checkmark covers the transition SHAPE only. Reading the limit as a NEW  $\perp'$  is a commitment; Snap.lean's NO-GO gauge holds T-SNAP in a model where nothing moves.
- Step 6 — DA-2 licenses  $\perp'$ : DA-2 (ZP-E) establishes that any state satisfying  $\forall x, S \vee x = x$  IS the  $\perp$  of its own lattice. That the Cauchy limit  $0 \in Q_2$  satisfies that condition is a HYPOTHESIS, never a result: `t_iz_complete` takes it as the argument `h_role`, about a terminal in a separate semilattice  $L'$ , and does not identify that terminal with the limit.  $Q_2$  carries no join at all — its ring structure supplies none with 0 as bottom (ZeroParadox/Valuation/ScaleBridge.lean) — so the condition is not statable there. Lean:

`t_iz_limit_is_new_null`, proved directly from `da2_bottom_characterization`. ✓ — the checkmark covers the ROLE identification only, and only inside  $L'$ . It is NOT a novelty witness and must not be cited as one; reading the occupant as the  $\perp$  of a SUCCESSOR instantiation is C-DA2.

## II. Proof Obligation Table

| Claim                                         | Source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | New axiom?                                     | Lean status                                                                                                        |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Chain is Cauchy in $(Q_2, \ \cdot\ _2)$       | The geometric norm bound. What the framework supplies to REACH it (via §Ia) is <code>IsStrictStateSequence</code> (T3 rides inside it — T3 alone is MONOTONE, and monotone does not reach this bound), <code>IsDepthChain</code> , the chain being nowhere zero, and the norm-valuation identity $\ x\ _2 = 2^{-v_2(x)}$ from ZP-B's valuation construction (Lean: <code>Padic.norm_eq_zpow_neg_valuation</code> ). NOT ZP-B T2, which is “every ball is clopen” — a separation result that says nothing about a valuation growing. <code>t_iz_cauchy</code> itself binds none of them, and no lattice at all. | Follows from existing structure — no new axiom | Lean: <code>t_iz_cauchy</code> ✓ (proved; carries <code>Classical.choice</code> , <code>Mathlib.p-adic</code> )    |
| $\ S(n)\ _2 \rightarrow 0$ (Cauchy limit = 0) | The geometric norm bound alone. Completeness of $Q_2$ is NOT consumed here: the limit is exhibited as 0 rather than obtained from a Cauchy criterion, so the proof runs through <code>squeeze_zero</code> and <code>tendsto_zero_iff_norm_tendsto_zero</code> , which hold in any normed group.                                                                                                                                                                                                                                                                                                                | Already in framework                           | Lean: <code>t_iz_cauchy</code> (composite of <code>t_iz_norm_tendsto_zero</code> + <code>t_iz_conv_zero</code> ) ✓ |

| Claim                                                                     | Source                                                                                                                                                                             | New axiom?                                                                 | Lean status                                                                                                                                                                                                          |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\sup v_2(S(n)) = \infty$                                                 | Strict ascent (IsStrictStateSequence) + IsDepthChain (the valuation tracks the depth index). NOT no-top: a chain can satisfy HasNoTop and IsDepthChain with its valuation BOUNDED. | Follows from strict ascent + T3 — no-top supplies the room, not the growth | Lean: <code>t_iz_valuation_unbounded</code> ✓ (proved; carries Classical.choice, Mathlib p-adic)                                                                                                                     |
| $v_2 \rightarrow \infty \implies K/ S  \rightarrow 1$                     | ZP-C D1 ( $P_0$ ) + L-INF + ZP-B (binary construction)                                                                                                                             | Informational context — not a proof dependency                             | Outside Lean scope. Not required: formal spine is Steps 1 + 6; DA-1 is closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement. Retained as motivational context. |
| $P_0$ fires DA-1                                                          | ZP-C D1 + DA-1 (ZP-E)                                                                                                                                                              | Already in framework                                                       | ZPE formal core: <code>da1_minimal_path</code> , DP-2 ✓                                                                                                                                                              |
| DA-1 fires T-SNAP                                                         | ZP-E T-SNAP                                                                                                                                                                        | Already in framework                                                       | Lean: <code>t_snap_derived</code> ✓ (axiom-free)                                                                                                                                                                     |
| The limit occupies the $\perp$ role (READING; $\perp'$ a further READING) | DA-2 (ZP-E)                                                                                                                                                                        | Role-recognition: already in framework. Occupancy and novelty: COMMITMENTS | Lean: <code>t_iz_limit_is_new_null</code> , <code>c_da2_novelty</code> ✓ — these prove role $\implies$ identity with the bottom ALREADY THERE. Neither is a novelty witness and neither may be cited as one.         |

### III. Lean Scope

The Lean file `SemilatticeInstance.lean` formalizes the formal spine of T-IZ: Step 1 (Cauchy convergence, § I) and Step 6 (DA-2 licensing of  $\perp'$ , § IV), which are carried together as a CONJUNCTION and are not joined to each other. The two steps sit on different footings, and the document states this honestly: Step 6 (`t_iz_limit_is_new_null`) is proved axiom-free — it is pure DA-2 structure. Step 1 (`t_iz_cauchy`) is proved sorry-free but carries Classical.choice, and NOT because of the analysis. Measured:  $(1 : \mathbb{Q}_{[2]}) \neq 0$  — no limit, no filter, no convergence lemma anywhere in the statement — already reports [propxet, Classical.choice, Quot.sound]. The dependence sits in Mathlib’s instance packaging, not in the metric: Polynomial (ZMod 5) is no kind of completion and reports the same footprint, while ZMod 5 reports [propxet, Quot.sound]. Mathlib’s analytic lemmas (`squeeze_zero`, the geometric-limit lemma, `tendsto_zero_iff_norm_tendsto_zero`)

carry it too, so the footprint cannot separate the two.  $\triangleleft$  A footprint is a fact about the PROOF, never about the type: the same statement  $(1 : \mathbb{Q}) \neq 0$  reports `Classical.choice` through `one_ne_zero` and `[propext, Quot.sound]` through `decide`.  $\triangleleft$  The corpus's choice-free valuation facts are NOT a counterexample here: `v2_bot` is `[propext]` because  $v2 : \mathbb{N} \rightarrow \mathbb{N}_\infty$  is stated over the `NATURAL`S, so that contrast is between two carriers and not between algebra and analysis. The footprint reports the library this development is built on. Where choice enters this corpus, and what is settled about it, is recorded at `ZeroParadox/AxiomProfile.lean` (the core is choice-free; `T-SNAP` depends on no axioms at all) and `ZeroParadox/Ordinal/SyntacticCollapse.lean` (a choice-free syntactic surrogate for the metric collapse, which states of itself that it settles the standing conjecture in neither direction). Steps 2–5 (the valuation-complexity bridge and `DA-1/T-SNAP` path) describe the original ZP-E informational argument and are retained as motivational context. `DA-1` is closed given `DP-2` (`da1_minimal_path`; `CLAIMS.md`, Tier 5); `ZP-K` witnesses its Path 1 (`da1_closed_concrete`:  $\perp$  is the unique Quine atom of `MachinePhase`). The `DP-2` commitment is not discharged by either. The theorems in `SemilatticeInstance.lean`, with their axiom footprints:

- `t_iz_limit_is_new_null`: anything satisfying the `DA-2`  $\perp$  role IS that lattice's  $\perp$  (the structural core of the snap-arc; axiom-free). The role property is its `HYPOTHESIS`; that the Cauchy limit satisfies it is not established here, and is not statable in  $\mathbb{Q}_2$ .
- `c_t_iz_null_balance`: a non-bottom state cannot satisfy the  $\perp$  role (choice-free, `[propext]`).
- `t_iz_cauchy`: the ascending chain converges to 0 (topological core; carries `Classical.choice` from `Mathlib`'s p-adic convergence, not a framework commitment).
- `t_iz_valuation_unbounded`:  $\sup v_2(S(n)) = \infty$  (carries `Classical.choice`, `Mathlib` p-adic).
- `t_iz_c3_compatible`: `C3` irreversibility is preserved — Cauchy sequences  $\neq$  continuous paths (carries `Classical.choice`, `Mathlib` p-adic).
- `t_iz_h_bound_from_depth_chain`: `h_bound` derived from `THREE` hypotheses, of which exactly one is a ZP-A lattice condition — `IsStrictStateSequence`, and it is a condition on the `DEPTH INDEX`. The other two live in  $\mathbb{Q}_2$ : that the chain is nowhere zero, and `IsDepthChain`, the `BRIDGE` saying the 2-adic valuation tracks that index. `IsDepthChain` takes no `ZPSemilattice` instance at all; `§Ib` records it as an undischarged modelling commitment, and it is what makes "the chain" in the lattice and "the chain" in  $\mathbb{Q}_2$  the same object. Closes the  $\|S_0\|$  factor gap between `§Ib` and `t_iz_complete` (optional transparency lemma; the analytic proof carries `Classical.choice`).
- `t_iz_complete_from_axioms`: `T-IZ` complete variant taking the three hypotheses above in place of a bare `h_bound`, and binding four more — a second semilattice `L'` with a `KleeneStructure` instance, a terminal, an  $\varepsilon_0'$ , and `h_role`, the role property `HANDED IN` and never derived. Auditable in one theorem, but its hypotheses are not all lattice conditions (carries `Classical.choice`; optional transparency variant; `t_iz_complete` is the canonical theorem).

Status: DERIVED THEOREM — formal spine: `t_iz_cauchy` (Step 1, proved; carries `Classical.choice` from `Mathlib` p-adic analysis) + `t_iz_limit_is_new_null` (Step 6, axiom-free via `DA-2`). These two steps are the formal spine, carried as a conjunction and not joined to one another. `t_iz_valuation_unbounded`, `c_t_iz_null_balance`, `t_iz_c3_compatible` also proved. Transparency variants: `t_iz_h_bound_from_depth_chain` + `t_iz_complete_from_axioms` name each hypothesis separately for reviewer auditability — one lattice condition (`IsStrictStateSequence`), one 2-adic depth bridge (`IsDepthChain`, a modelling commitment), one nonvanishing condition, and for the complete variant `h_role` handed in — `t_iz_complete` is the canonical theorem. Steps 2–5 (valuation-complexity bridge +

DA-1/T-SNAP) are informational context — DA-1 is closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement, no Kolmogorov complexity required. No new axioms. ✓

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## Section IV: Compatibility with the Irreversibility Results

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T-IZ does not violate any irreversibility result in the framework. The inside approach is not a reversal — it is a structurally different operation. Each irreversibility result governs a specific structure; T-IZ uses a different structure not governed by any of them.

### I. ZP-A R1 — No Subtraction

R1 states that the join-semilattice  $(L, \vee, \perp)$  has no subtraction operator: for any  $x, y \in L$  with  $x < y$ , there is no  $z$  such that  $y \vee z = x$ . This closes the algebraic door to reversal.

T-IZ does not use subtraction. The chain never joins "downward." Every step is a join operation  $S_{n+1} = S_n \vee \alpha_n$  for some  $\alpha_n \geq 0$  (ZP-A T3 monotonicity). The approach to 0 in  $Q_2$  is not a join toward 0 — it is a Cauchy sequence whose 2-adic norm tends to 0. The chain never subtracts, so R1 is not violated. R1 plays no further part in T-IZ: what makes the valuation grow without bound is strict ascent, and what leaves the chain somewhere to go is the order property HasNoTop (ZeroParadox/Order/Lattice.lean), which is a different proposition.

### II. ZP-B C3 — No Continuous Path to Zero

C3 states: there is no continuous path  $\gamma : [0,1] \rightarrow Q_2$  with  $\gamma(0) = x \neq 0$  and  $\gamma(1) = 0$ . This closes the topological door to reversal via continuous motion.

T-IZ uses Cauchy sequence convergence, not a continuous path. A Cauchy sequence  $(S_n)_{n \in \mathbb{N}}$  tending to 0 is a countable sequence of discrete points. It is not a continuous function  $[0,1] \rightarrow Q_2$ . These are distinct mathematical structures. C3's universal quantifier ranges over continuous functions; T-IZ's convergence is a statement about countable sequences. The two results do not conflict.

Lean: `t_iz_c3_compatible` (SemilatticeInstance.lean) proves this directly: the statement of C3 (`c3_irreversible` from ZPB) holds without modification alongside T-IZ. C3 blocks continuous paths; T-IZ uses Cauchy sequences. They govern different structures. ✓

### III. ZP-G AX-G2 — No Morphism to the Initial Object

AX-G2 states that in the categorical structure  $C$ ,  $\text{hom}(X, 0) = \emptyset$  for  $X \neq 0$ : no morphism within  $C$  leads back to the initial object. This closes the categorical door to reversal.

T-IZ is not a morphism within  $C$ . The transition to  $\perp'$  is not an arrow in the category  $C$  of the current instantiation. It is the termination of  $C$  and the opening of  $C'$ . AX-G2 quantifies only over morphisms within a single category; it has nothing to say about the transition between categories. The categorical structure is preserved intact within each instantiation.

## IV. Summary

The irreversibility results and T-IZ are not in tension. They describe different things. Irreversibility (R1, C3, AX-G2) governs motion within an instantiation branch: no algebraic subtraction, no continuous topological

return, no categorical reversal. T-IZ governs what happens at the branch's ordinal limit: the chain converges, by Cauchy convergence, to an occupant of the bottom role — a structure that none of the irreversibility results governs or addresses. Reading that occupant as a successor null is C-DA2.

The inside approach is not a violation of irreversibility. It is the discovery of a structure that irreversibility does not reach. Three doors to zero are closed (R1, C3, AX-G2). T-IZ uses a fourth passage — Cauchy sequence convergence — that none of the three irreversibility theorems govern.

## Section V: Framework Closure — OQ-E2, the Null Balance, and the Complete Cycle

### I. Resolution of OQ-E2

OQ-E2 (Cardinality-Semilattice Correspondence) has been open since the initial ZP-E release. It asks: do specific semilattice structures correspond to specific cardinality regimes, and can the framework make predictions about which instantiations satisfy CH?

T-IZ provides the path to closing OQ-E2 as follows. The ascending chain  $(S_n)_{n < \omega}$  is indexed by  $\omega$  — the first infinite ordinal. This indexing is forced, not chosen: the binary alphabet gives a countable state space;  $Q_2$  is a separable metric space; surprisal  $I(n) = n$  grows by integer steps (ZP-C D4). Every component of the framework that generates an ordinal index generates a countable one. The state sequence is necessarily indexed by  $\omega$ , not  $\omega_1$  or any uncountable ordinal.

This pins the ordinal depth of each instantiation to  $\omega$ . OQ-E2's perspective-relative cardinality (DA-3) is then resolved as follows: internal observers see a proper initial segment of  $\omega$  (finite); external observers see all of  $\omega$ . The perspective-relativity is ordinal, not set-theoretically free — it is the difference between a finite position in the chain and the view of the full chain from outside. The cardinality of the fan at each node is determined by the countable substrate.

OQ-E2 status after T-IZ: **PARTIALLY CLOSED**. The ordinal indexing  $\Omega = \omega$  is forced by the countable binary substrate (ZP-C D4, ZP-B  $Q_2$  separability, binary alphabet). Internal/external perspective relativity is ordinal, not set-theoretically free. Formal connection between specific semilattice structures and specific CH instances remains deferred — that is the remaining open question in OQ-E2. ✓ (partial)

### II. The Null Balance

The null balance  $0 + x + (-x) = 0$  is an arithmetic **READING** of the complete cycle of an instantiation branch, never a theorem of the structure: a ZPSemilattice carries join and  $\perp$  and no additive inverse, so the identity is not statable there at all. As a reading it describes the cycle of an instantiation branch: it begins at  $\perp$  (0), generates  $\varepsilon_0$  and successors  $(+x)$ , and at the ordinal limit returns to 0  $(-x)$ , read as the bottom role and then as  $\perp'$ . The three terms are strung across  $\omega$  state changes.

T-IZ establishes that this balance is exact and derived. "Balance" here is not subtraction in  $(L, \vee, \perp)$  — R1 prohibits that. It is the completion of an instantiation branch: the closing of L and the emergence of  $L'$ . Every instantiation begins at its  $\perp$ , ascends for  $\omega$  state changes under T3 (monotonicity), and at the limit converges to 0. Reading that limit as an occupant of the bottom role is a commitment; on it, T-IZ + T-SNAP + DA-2 close the branch. The **CONVERGENCE** holds in every instantiation as a theorem; that the occupant is a

DISTINCT  $\perp'$  rather than the  $\perp$  it began at is the commitment, and in the 2-adic realization the arc returns to the same 0 (snap\_arc\_z2\_loop).

Null Balance (role-RECOGNITION derived; occupancy and novelty COMMITTED): For every ascending chain  $(S_n)_{n < \omega}$  in the Zero Paradox framework with  $S_0 = \perp$  (CC-1),  $v_2(S_n) \rightarrow \infty$  (forced by strict ascent), and  $v_2(S(n)) \geq n$  (derived via h\_strict\_from\_r1\_t3 given IsDepthChain and IsStrictStateSequence — R-IZ-A closed): the chain converges to 0. Reading that limit as an occupant of the bottom role, and that occupant as a successor  $\perp'$ , are the two commitments — neither is carried by T-IZ. The balance  $0 + x + (-x) = 0$  is a READING of that cycle and not a theorem — a ZPSemilattice has no additive inverse, so it is not statable in the structure T-IZ is about — where  $x$  represents  $\omega$  state changes under T3, and  $(-x)$  represents the return to 0. No new axioms required for the convergence; the readings are commitments, not axioms.

### III. The Complete Cycle

The Zero Paradox now describes a complete cycle. In the original T-SNAP picture, the framework had a beginning (T-SNAP:  $\perp \rightarrow \epsilon_0$  — shape derived, occurrence committed to) but no clear closing structure. T-IZ provides the closure:

- T-SNAP: From  $\perp$ , existence emerges — the shape of that emergence is derived and its occurrence is committed to, not proved. The Binary Snap  $\perp \rightarrow \epsilon_0$  is irreversible. This is the opening of the branch.
- T3 (Monotonicity): The state sequence ascends without interruption. Each step adds informational content irreversibly. The chain climbs.
- No Top: There is always a strictly greater element, so no chain halts for want of anywhere to go. It supplies the POSSIBILITY of ascent; IsStrictStateSequence is its OCCURRENCE, and that is the hypothesis T-IZ actually consumes. Not ZP-A's R1, which is no-subtraction.
- T-IZ: The chain's unbounded forward motion generates the conditions for a null at the ordinal limit  $\omega$ . DA-1 fires; T-SNAP fires again; the limit is identified as  $\perp'$  by its role. Calling that role-occupant a SUCCESSOR rather than the same  $\perp$  is the commitment. This is the closing of the branch.
- DA-2 (Instantiation Succession):  $\perp'$  becomes the foundation of the next instantiation. The tree extends. The cycle repeats.

The framework is a closed system. The formal spine of T-IZ takes  $v_2(S(n)) \geq n$  as a condition derived from IsStrictStateSequence via the IsDepthChain modeling commitment (h\_strict\_from\_r1\_t3, §Ib — R-IZ-A closed). Given that condition,  $\perp$  is not just the bottom of the lattice — it is the attractor of the chain's own unbounded forward motion. The framework does not end with emergence. Emergence is the opening of a cycle that is self-closing by structure.

#### Key Result

T-SNAP:  $\perp \rightarrow \epsilon_0$ , shape derived and occurrence committed (existence emerges from null).

T-IZ:  $(\perp, \epsilon_0, \epsilon_1, \dots) \rightarrow 0$  at  $\omega$ , ROLE-RECOGNITION derived; OCCUPANCY and NOVELTY committed. What is proved is the 2-adic convergence (t\_iz\_cauchy) and, separately, that anything playing the bottom role IS the bottom (t\_iz\_limit\_is\_new\_null). That the limit is a thing playing that role is a commitment, not a theorem — the role condition is not statable in  $Q_2$ ; that the bottom so reached is a NEW one is a further commitment.

Framework closure: the Zero Paradox is a closed system. Strict valuation growth is Lean-derived from `IsDepthChain + IsStrictStateSequence (h_strict_from_r1_t3)`. Emergence and return are derived as far as the CONVERGENCE and the role-recognition implication; that the limit is the role's occupant is committed, and their novelty is committed, on the same footing as T-SNAP's occurrence. No new axioms required beyond AX-B1, AX-G1, AX-G2.

## Open Items Register

| Item                                          | Status                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-IZ: Inside Zero Theorem                     | CONVERGENCE + ROLE-RECOGNITION DERIVED — occupancy and novelty COMMITTED | Every maximal ascending chain that strictly ascends at every step converges to 0 in $Q_2$ ; reading that limit as the occupant of the bottom role, and that occupant as a successor null, are commitments. Formal spine: Step 1 ( <code>t_iz_cauchy</code> , carries Mathlib p-adic <code>Classical.choice</code> ) + Step 6 ( <code>t_iz_limit_is_new_null</code> , axiom-free via DA-2). Steps 2–5 are informational context — original ZP-E path; DA-1 is closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement. No new axioms required. |
| OQ-E2: Cardinality-semilattice correspondence | PARTIALLY CLOSED — $\Omega = \omega$ forced                              | Ordinal indexing $\Omega = \omega$ forced by countable binary substrate (ZP-C D4, $Q_2$ separability). Internal/external perspective relativity is ordinal, not set-theoretically free. Formal connection to specific CH instances: still open — deferred to future work.                                                                                                                                                                                                                                                                                                                        |
| Null balance: $0 + x + (-x) = 0$              | CLOSED for ROLE-RECOGNITION — occupancy and novelty COMMITTED            | The convergence half is derived from T-IZ: every branch starts at $\perp$ , ascends for $\omega$ state changes (T3), and at the limit converges to 0. Reading that limit as an occupant of the bottom role is a COMMITMENT, not a consequence — the join-identity is not statable in $Q_2$ . On that reading, DA-2 identifies the occupant with the $\perp$ already there (T-IZ + T-SNAP + DA-2). " $-x$ " is not subtraction in $L$ — it is the return to that role by forward motion. That the occupant is a NEW $\perp$ is the commitment, not the theorem.                                   |
| Valuation-complexity bridge                   | CONTEXTUAL — informational layer                                         | Original ZP-E path connecting 2-adic depth to Kolmogorov complexity and DA-1 Path 3. Not a proof dependency for T-IZ: formal spine is Steps 1 + 6 (Step 6 axiom-free; Step 1 carries Mathlib p-adic choice); DA-1 is closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement. Retained as motivational context documenting convergence of the topological and informational layers at $P_0$ . Outside Lean scope (Kolmogorov complexity absent from standard proof libraries) — but no longer load-bearing.                                   |

| Item                        | Status                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-IZ Lean sorry fill        | CLOSED — ZeroParadox/Valuation/SemilatticeInstance.lean                                           | t_iz_norm_tendsto_zero and t_iz_conv_zero filled; t_iz_cauchy proved (carries Mathlib p-adic Classical.choice). All SemilatticeInstance.lean theorems compile with no sorry. Axiom footprint: standard foundational axioms only (propext, Quot.sound, and Classical.choice from Mathlib p-adic analysis on the convergence theorems; the DA-2 step is axiom-free).                                                     |
| AX-1: Binary Snap Causality | RETIRED — shape proved as T-SNAP (ZP-E); the snap occurs given the occurrence commitment and DA-1 | AX-1 (Binary Snap Causality) is retired. Its content was split in two: the shape of the snap is proved, as Theorem T-SNAP, and that the snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). t_snap_holds_but_nothing_moves shows T-SNAP does not carry it. T-SNAP is derived in ZP-E. T-IZ extends T-SNAP to the ordinal limit. |
| Remaining axioms            | INTENTIONAL — AX-B1, AX-G1, AX-G2                                                                 | These are the named axioms. The occurrence commitment is not on this list because it is a commitment, not a named axiom. T-IZ requires no additions.                                                                                                                                                                                                                                                                   |

## Traceability Register

| Claim                           | Grounded In                                                                                                                                                                                                                                                                                  | Bridge Axiom ? | Status                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-IZ: Inside Zero               | What the chain CONSUMES: IsStrictStateSequence, IsDepthChain, the chain being nowhere zero, and ZP-B's norm-valuation identity. HasNoTop supplies the room and is consumed by nothing here. Also ZP-B T5/C3 for the irreversibility comparison only; ZP-C L-INF, D1; ZP-E DA-1, T-SNAP, DA-2 | None           | Role derived — T-IZ ✓ (formal spine Steps 1+6: Step 6 axiom-free, Step 1 carries Mathlib p-adic choice; bridge: contextual). Novelty of the successor bottom: COMMITMENT, not proved. |
| Null Balance $0 + x + (-x) = 0$ | T-IZ + T-SNAP + DA-2 (ZP-E)                                                                                                                                                                                                                                                                  | None           | Role derived — consequence of T-IZ. Exact, not approximated. Novelty: commitment.                                                                                                     |
| OQ-E2 partial closure           | ZP-C D4 (binary alphabet, $I(n)=n$ ); ZP-B ( $Q_2$ separable); T-IZ ( $\Omega = \omega$ )                                                                                                                                                                                                    | None           | Partially closed — $\Omega = \omega$ forced by countable substrate; CH connection deferred.                                                                                           |
| t_iz_valuation_unbounded (Lean) | int_strict_mono_ge (induction on $\mathbb{Z}$ ); omega (integer arithmetic)                                                                                                                                                                                                                  | None           | Lean: proved ✓ — standard foundational axioms only. Formalises " $\sup v_2(S(n)) = \infty$ " — proof obligation table row 3.                                                          |

| Claim                                | Grounded In                                                                                                                                     | Bridge Axiom ? | Status                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t_iz_cauchy (Lean)                   | ZP-B ( $\mathbb{Q}_2$ normed field); geometric tendsto; Mathlib.Analysis.SpecificLimits.Basic                                                   | None           | Lean: proved ✓ (t_iz_norm_tendsto_zero, t_iz_conv_zero filled; carries Mathlib p-adic Classical.choice)                                                                                                                                                                                                                                                                                       |
| t_iz_limit_is_new_null (Lean)        | da2_bottom_characterization                                                                                                                     | None           | Lean: proved ✓ (direct delegation to DA-2)                                                                                                                                                                                                                                                                                                                                                    |
| c_t_iz_null_balance (Lean)           | c_da2_novelty                                                                                                                                   | None           | Lean: proved ✓ (direct delegation to C-DA2)                                                                                                                                                                                                                                                                                                                                                   |
| t_iz_c3_compatible (Lean)            | c3_irreversible                                                                                                                                 | None           | Lean: proved ✓ (C3 holds unmodified; Cauchy sequences $\neq$ continuous paths)                                                                                                                                                                                                                                                                                                                |
| Valuation-complexity bridge          | ZP-C D1, L-INF; ZP-B T2; AIT (standard)                                                                                                         | N/A            | Informational context — not load-bearing. DA-1 is closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement. Outside Lean scope (Kolmogorov complexity absent from standard proof libraries).                                                                                                                                                |
| t_iz_h_bound_from_depth_chain (Lean) | h_strict_from_r1_t3 (§Ib); t_iz_r1_t3_geometric_bound; Padic.norm_eq_zpow_neg_valuation (depths $0 : \mathbb{N} \Rightarrow \ S_0\ _2 \leq 1$ ) | None           | Lean: proved ✓ — optional transparency lemma. Derives h_bound from IsStrictStateSequence (the one lattice condition), IsDepthChain (the 2-adic bridge, binding no semilattice) and the chain being nowhere zero, closing the $\ S_0\ $ factor gap between §Ib and t_iz_complete.                                                                                                              |
| t_iz_complete_from_axioms (Lean)     | t_iz_h_bound_from_depth_chain; t_iz_complete                                                                                                    | None           | Lean: proved ✓ — optional transparency variant. Replaces h_bound with three hypotheses, of which only IsStrictStateSequence is a ZP-A lattice condition; IsDepthChain is the bridge to the 2-adic valuation and binds no semilattice, and the chain must be nowhere zero. It still carries h_role explicitly, which is the hypothesis nothing here grounds. Canonical theorem: t_iz_complete. |

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End of ZP-I | Theorem T-IZ: Inside Zero | R-IZ-A closed: strict valuation growth derived from IsDepthChain + IsStrictStateSequence (h\_strict\_from\_r1\_t3, §Ib) | Framework closure: no construction-level hypothesis required | Formal spine: Step 6 axiom-free (t\_iz\_limit\_is\_new\_null), Step 1 carries Mathlib p-adic Classical.choice (t\_iz\_cauchy) | Valuation-complexity bridge: informational context, not load-bearing | DA-1 closed given DP-2 (ZP-E), with ZP-K witnessing Path 1 and carrying Path 3 as a KleeneStructure requirement | Remaining axioms: AX-B1, AX-G1, AX-G2 | No new axioms required